

Midterm test, Term 1 — Speed, energy and work: solutions, versions A, B and C

Total 55 marks. The parts marked  are worth 33 marks, 60%: that is the pass mark. Marks are given part by part. Each exercise ends with the common mistakes and the wrong numbers they produce.

Version A

Exercise 1 — Multiple choice (8 marks, 8 🍉)

Solution — 1-B, 2-C, 3-D, 4-B, 5-A, 6-C, 7-D, 8-A.

1. A battery stores chemical energy.
2. $v = s/t = 240/3.0 = 80 \text{ km/h}$.
3. The gradient of the graph is the speed.
4. At the top the speed is zero, so $E_k = 0$ and E_p is maximum.
5. $E_k = \frac{1}{2}mv^2$ is proportional to m .
6. The force moves the sledge through a distance. In the other cases nothing moves.
7. $W = Fd = 5.0 \times 3.0 = 15 \text{ J}$. The answer 15 N has the wrong unit.
8. $500 - 120 = 380 \text{ J}$. Adding gives 620; 120 is the thermal energy; 500 forgets the transfer.

Exercise 2 — True or false (4 marks, 4 🍉)

Solution — (a) F, (b) T, (c) F, (d) T.

(a) $E_k \propto v^2$: it becomes four times bigger. (b) Principle of conservation of energy. (c) No movement, no distance, no work. (d) $W = Fd$.

Exercise 3 — Energy stores (3 marks, 3 🍉)

Solution — A girl releases a stretched bowstring and the arrow flies off: elastic (strain) $\rightarrow$ kinetic; A cyclist freewheels down a hill without pedalling: gravitational potential $\rightarrow$ kinetic; A student runs up a flight of stairs: chemical $\rightarrow$ gravitational potential. Award the mark only if both stores are right.

Exercise 4 — Two travellers (8 marks, 2 🍉)

Solution — (a) 🍉 [2] The cyclist is at rest for 0.5 h (horizontal line from 0 to 0.5 h). The graphs cross at $t = 1.5 \text{ h}$, 15 km from P. 1 mark each.

(b) [2] While moving, from 0.5 h to 2.5 h, the distance from P changes by 30 km in 2 h. The line goes down because the cyclist is getting closer to P; its speed is the size of the gradient.

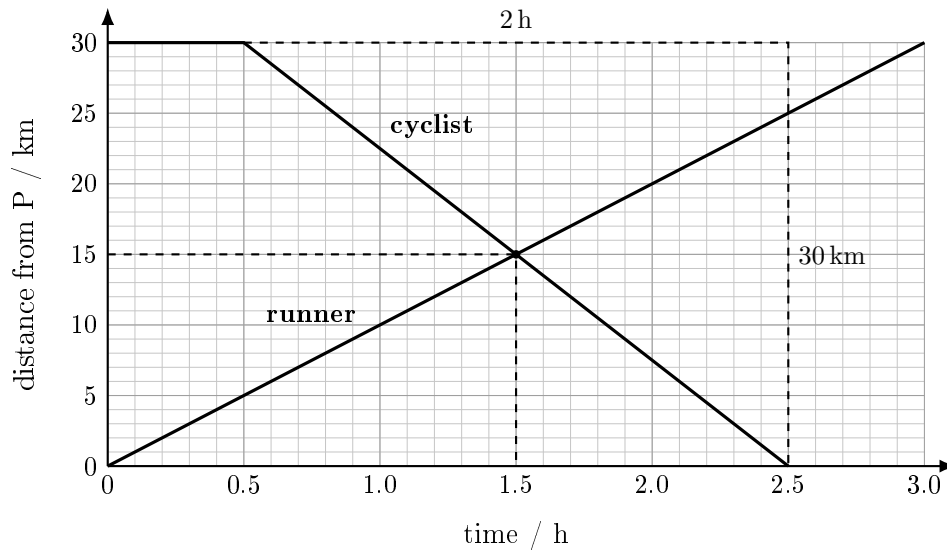
$$v = \frac{s}{t} = \frac{30 \text{ km}}{2 \text{ h}} = 15 \text{ km/h}$$

(c) [2] The cyclist covers 30 km in the whole time 2.5 h, including the rest.

$$v_{\text{av}} = \frac{30 \text{ km}}{2.5 \text{ h}} = 12 \text{ km/h}$$

(d) [2] The runner covers 30 km in 3 h, so $v = 10 \text{ km/h}$.

$$v = \frac{10\,000 \text{ m}}{3600 \text{ s}} \approx 2.78 \text{ m/s}$$



Check and common mistakes — (a) Writing 1.5 h for the rest (that is the time of the meeting) or giving the meeting point in hours instead of kilometres. (b) Using the whole time 2.5 h gives 12 km/h: the rest is not moving time. Dividing time by distance gives 0.067: the unit would be h/km. (c) Using 2 h gives 15 km/h: the rest counts in the average. Averaging 0 and 15 gives 7.5 km/h: wrong. (d) Multiplying by 3.6 instead of dividing gives 36 m/s: not a possible speed for a runner. Check: $2.78 \text{ m/s} \times 3.6 \approx 10 \text{ km/h}$.

Exercise 5 — The half-pipe (8 marks, 4 ⚽)

Solution — (a) ⚽ [2] She rises to 3.0 m above C, the height of A (1 mark). At A she has only gravitational potential energy; no energy is lost, so at the highest point, where she stops for an instant, she again has only gravitational potential energy and the total is the same (1 mark). Accept 1.8 m above E if it is clearly stated as the height above E.

(b) ⚽ [2]

$$mgh_A = mgh_E + \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2g(h_A - h_E)}$$

1 mark for the energy balance, 1 for the final expression. The mass cancels.

(c) [2]

$$\Delta E_p = mg\Delta h = (55)(10)(3.0 - 1.2) = (55)(10)(1.8) = 990 \text{ J}$$

(d) [2] The potential energy lost becomes kinetic energy:

$$\begin{aligned} \frac{1}{2}mv^2 &= 990 \Rightarrow v^2 = \frac{2 \times 990}{55} = 36 \\ v &= \sqrt{36} = 6.0 \text{ m/s} \end{aligned}$$

Check and common mistakes — (a) 1.8 m is the height above the lip, not above C. (c) Using h_A alone gives 1650 J; using h_E alone gives 660 J: the energy lost depends on the difference. Forgetting g gives 99 J. (d) Forgetting the $\frac{1}{2}$ gives $v^2 = 18$, $v \approx 4.2 \text{ m/s}$. Forgetting the square root gives 36 m/s. Check with (b): $\sqrt{2 \times 10 \times 1.8} = 6.0 \text{ m/s}$.

Exercise 6 — Lifting and carrying a box (8 marks, 4 ⚽)

Solution — (a) ⚽ [2] $W = Fd$ (work done by the hands, with d the distance moved in the direction of the force) and $\Delta E_p = mg\Delta h$. 1 mark each.

(b) [2]

$$W = Fd = (200)(1.2) = 240 \text{ J}$$
$$\Delta E_p = mg\Delta h = (15)(10)(1.2) = 180 \text{ J}$$

(c) [1] The difference is $240 - 180 = 60 \text{ J}$. It is the kinetic energy of the box, which is still moving at the height of a shelf.

(d) [1] 0 J.

(e) [2] The force of the hands is vertical and the box moves horizontally: the box does not move in the direction of the force, so $d = 0$ in $W = Fd$ and $W = 0$ (1 mark). The box gains no energy: its height and speed do not change (1 mark). He still uses chemical energy in the muscles, but that is not work done on the box.

Check and common mistakes — (b) Using the weight $mg = 150 \text{ N}$ instead of the force of 200 N gives $150 \times 1.2 = 180 \text{ J}$, equal to the gain in E_p : the difference disappears. Forgetting g in ΔE_p gives $15 \times 1.2 = 18 \text{ J}$. (d) Using 8.0 m gives $150 \times 8.0 = 1200 \text{ J}$: that is the distance moved, but not in the direction of the force. (e) Answers like “carrying costs no energy” are wrong: the muscles transfer chemical energy to thermal energy even when no work is done on the box.

Exercise 7 — A car brakes (6 marks, 2)

Solution — (a) [1] Thermal (internal) energy of the brakes, the tyres and the road.

(b) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(1200)(15)^2 = 135\,000 \text{ J}$$

(c) [2] The work done by the brakes equals the kinetic energy lost: $Fd = E_k$ (1 mark).

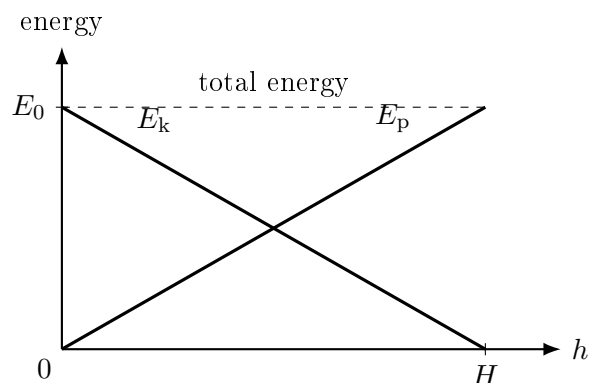
$$F = \frac{E_k}{d} = \frac{135\,000}{25} = 5400 \text{ N}$$

(d) [1] Twice as long: the kinetic energy doubles with the mass ($E_k \propto m$) and, with the same force, $d \propto E_k$. Check: $d = 50 \text{ m}$.

Check and common mistakes — (b) Forgetting the square gives 9000 J . Forgetting the $\frac{1}{2}$ gives $270\,000 \text{ J}$. (c) With 9000 J from (b) the force would be 360 N . Dividing by the mass gives 112.5 : that is not a force (the unit would be J/kg). (d) “Four times as long” is the answer for a doubled speed, not a doubled mass.

Exercise 8 — A stone thrown upwards (10 marks, 6)

Solution — (a) [2]



The kinetic energy decreases linearly from E_0 to 0 (1 mark). The potential energy increases linearly from 0 to E_0 (1 mark). The two lines cross at $H/2$ and the total stays constant.

(b)  [2]

$$\frac{1}{2}mv^2 = mgH \quad \Rightarrow \quad H = \frac{v^2}{2g}$$

1 mark for the equality, 1 for the final expression. The mass cancels.

(c) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(0.2)(12)^2 = 14.4 \text{ J}$$

(d) [2]

$$H = \frac{E_k}{mg} = \frac{14.4}{(0.2)(10)} = 7.2 \text{ m}$$

(e)  [2] Four times as big (1 mark): $H \propto v^2$, so doubling v multiplies H by 4 (1 mark). Check: $H = 28.8 \text{ m}$.

Check and common mistakes — (c) Forgetting the square gives 1.2 J. (d) Forgetting g gives 72 m. Check with (b): $H = v^2/2g = 12^2/20 = 7.2 \text{ m}$.

Version B

Exercise 1 — Multiple choice (8 marks, 8 ⚽)

Solution — 1-C, 2-A, 3-B, 4-D, 5-C, 6-B, 7-A, 8-D.

1. The ball is at rest at a height: gravitational potential energy.
2. $v = s/t = 45/1.5 = 30$ km/h.
3. The distance does not change with time.
4. Conservation of energy.
5. $E_k \propto v^2$ and $3^2 = 9$.
6. The force moves the suitcase through a distance.
7. $W = Fd = 20 \times 0.50 = 10$ J. 40 comes from dividing F by d ; 0.025 from dividing d by F .
8. Conservation of energy: all the potential energy lost becomes kinetic energy.

Exercise 2 — True or false (4 marks, 4 ⚽)

Solution — (a) F, (b) T, (c) T, (d) F.

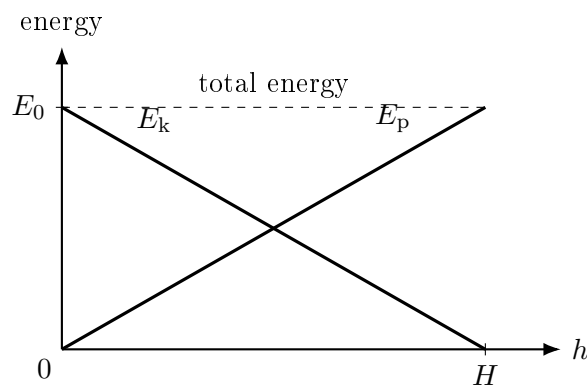
(a) It shows an object at rest. (b) The height does not change, so $\Delta E_p = 0$. (c) $E_k = \frac{1}{2}mv^2$. (d) Work is measured in joules.

Exercise 3 — Energy stores (3 marks, 3 ⚽)

Solution — A trolley rolls down a ramp: gravitational potential $\rightarrow$ kinetic; A stretched elastic band shoots a paper ball: elastic (strain) $\rightarrow$ kinetic; A crane lifts a load, using fuel: chemical $\rightarrow$ gravitational potential. Award the mark only if both stores are right.

Exercise 4 — A ball thrown upwards (10 marks, 6 ⚽)

Solution — (a) ⚽ [2]



The kinetic energy decreases linearly from E_0 to 0 (1 mark). The potential energy increases linearly from 0 to E_0 (1 mark). The two lines cross at $H/2$ and the total stays constant.

(b) ⚽ [2]

$$\frac{1}{2}mv^2 = mgH \Rightarrow H = \frac{v^2}{2g}$$

1 mark for the equality, 1 for the final expression. The mass cancels.

(c) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(0.4)(15)^2 = 45 \text{ J}$$

(d) [2]

$$H = \frac{E_k}{mg} = \frac{45}{(0.4)(10)} = 11.25 \text{ m}$$

(e) [2] Four times as big (1 mark): $H \propto v^2$, so doubling v multiplies H by 4 (1 mark). Check: $H = 45 \text{ m}$.

Check and common mistakes — (c) Forgetting the square gives 3 J. (d) Forgetting g gives 112.5 m. Check with (b): $H = v^2/2g = 15^2/20 = 11.25 \text{ m}$.

Exercise 5 — Two travellers (8 marks, 2)

Solution — (a) [2] The scooter is at rest for 0.5 h (horizontal line from 0 to 0.5 h). The graphs cross at $t = 1.5 \text{ h}$, 30 km from P. 1 mark each.

(b) [2] While moving, from 0.5 h to 2.5 h, the distance from P changes by 60 km in 2 h. The line goes up because the scooter is moving away from P.

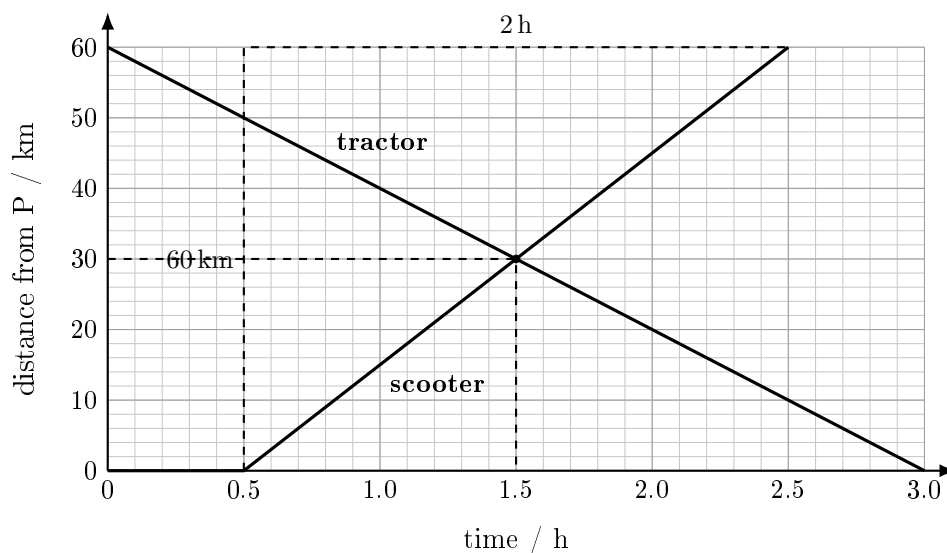
$$v = \frac{s}{t} = \frac{60 \text{ km}}{2 \text{ h}} = 30 \text{ km/h}$$

(c) [2] The scooter covers 60 km in the whole time 2.5 h, including the rest.

$$v_{\text{av}} = \frac{60 \text{ km}}{2.5 \text{ h}} = 24 \text{ km/h}$$

(d) [2] The tractor covers 60 km in 3 h, so $v = 20 \text{ km/h}$.

$$v = \frac{20\,000 \text{ m}}{3600 \text{ s}} \approx 5.56 \text{ m/s}$$



Check and common mistakes — (a) Writing 1.5 h for the rest (that is the time of the meeting) or giving the meeting point in hours instead of kilometres. (b) Using the whole time 2.5 h gives 24 km/h: the rest is not moving time. Dividing time by distance gives 0.033: the unit would be h/km. (c) Using 2 h gives 30 km/h: the rest counts in the average. Averaging 0 and 30 gives 15 km/h: wrong. (d) Multiplying by 3.6 instead of dividing gives 72 m/s: not a possible speed for a tractor. Check: $5.56 \text{ m/s} \times 3.6 \approx 20 \text{ km/h}$.

Exercise 6 — A van brakes (6 marks, 2)

Solution — (a) [1] Thermal (internal) energy of the brakes, the tyres and the road.

(b) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(2000)(12)^2 = 144\,000 \text{ J}$$

(c) [2] The work done by the brakes equals the kinetic energy lost: $Fd = E_k$ (1 mark).

$$F = \frac{E_k}{d} = \frac{144\,000}{30} = 4800 \text{ N}$$

(d) [1] One quarter as long: the kinetic energy becomes one quarter ($E_k \propto v^2$) and, with the same force, $d \propto E_k$. Check: $d = 7.5 \text{ m}$.

Check and common mistakes — (b) Forgetting the square gives 12 000 J. Forgetting the $\frac{1}{2}$ gives 288 000 J. (c) With 12 000 J from (b) the force would be 400 N. Dividing by the mass gives 72: that is not a force (the unit would be J/kg). (d) “Half as long” forgets that E_k depends on the square of the speed.

Exercise 7 — The half-pipe (8 marks, 4 [10])

Solution — (a) [2] He rises to 2.6 m above C, the height of A (1 mark). At A he has only gravitational potential energy; no energy is lost, so at the highest point, where he stops for an instant, he again has only gravitational potential energy and the total is the same (1 mark). Accept 0.8 m above E if it is clearly stated as the height above E.

(b) [2]

$$mgh_A = mgh_E + \frac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{2g(h_A - h_E)}$$

1 mark for the energy balance, 1 for the final expression. The mass cancels.

(c) [2]

$$\Delta E_p = mg\Delta h = (60)(10)(2.6 - 1.8) = (60)(10)(0.8) = 480 \text{ J}$$

(d) [2] The potential energy lost becomes kinetic energy:

$$\begin{aligned} \frac{1}{2}mv^2 &= 480 \quad \Rightarrow \quad v^2 = \frac{2 \times 480}{60} = 16 \\ v &= \sqrt{16} = 4.0 \text{ m/s} \end{aligned}$$

Check and common mistakes — (a) 0.8 m is the height above the lip, not above C. (c) Using h_A alone gives 1560 J; using h_E alone gives 1080 J: the energy lost depends on the difference. Forgetting g gives 48 J. (d) Forgetting the $\frac{1}{2}$ gives $v^2 = 8$, $v \approx 2.8 \text{ m/s}$. Forgetting the square root gives 16 m/s. Check with (b): $\sqrt{2 \times 10 \times 0.8} = 4.0 \text{ m/s}$.

Exercise 8 — Lifting and carrying a box (8 marks, 4 [10])

Solution — (a) [2] $W = Fd$ (work done by the hands, with d the distance moved in the direction of the force) and $\Delta E_p = mg\Delta h$. 1 mark each.

(b) [2]

$$\begin{aligned} W &= Fd = (300)(1.6) = 480 \text{ J} \\ \Delta E_p &= mg\Delta h = (25)(10)(1.6) = 400 \text{ J} \end{aligned}$$

(c) [1] The difference is $480 - 400 = 80 \text{ J}$. It is the kinetic energy of the box, which is still moving at the height of a high shelf.

(d) [1] 0 J.

(e) [2] The force of the hands is vertical and the box moves horizontally: the box does not move in the direction of the force, so $d = 0$ in $W = Fd$ and $W = 0$ (1 mark). The box gains no energy: its

height and speed do not change (1 mark). He still uses chemical energy in the muscles, but that is not work done on the box.

Check and common mistakes — (b) Using the weight $mg = 250\text{ N}$ instead of the force of 300 N gives $250 \times 1.6 = 400\text{ J}$, equal to the gain in E_p : the difference disappears. Forgetting g in ΔE_p gives $25 \times 1.6 = 40\text{ J}$. (d) Using 12.0 m gives $250 \times 12.0 = 3000\text{ J}$: that is the distance moved, but not in the direction of the force. (e) Answers like “carrying costs no energy” are wrong: the muscles transfer chemical energy to thermal energy even when no work is done on the box.

Version C

Exercise 1 — True or false (4 marks, 4 🍷)

Solution — (a) F, (b) T, (c) F, (d) T.

(a) It is transferred to the kinetic store: energy is conserved. (b) $\Delta E_p = mg\Delta h$. (c) You divide by 3.6. (d) That is the definition of work.

Exercise 2 — Multiple choice (8 marks, 8 🍷)

Solution — 1-D, 2-B, 3-A, 4-C, 5-D, 6-B, 7-C, 8-A.

1. Fuels store chemical energy.
2. $v = s/t = 400/50 = 8.0 \text{ m/s}$.
3. A constant gradient means a constant speed.
4. Friction transfers energy to the thermal store; the total is conserved.
5. $E_k \propto v^2$ and $(1/2)^2 = 1/4$.
6. Nothing moves, so there is no distance and no work.
7. $W = Fd = 30 \times 0.20 = 6.0 \text{ J}$. 150 comes from dividing F by d ; 0.0067 from dividing d by F .
8. $\Delta E_p = mg\Delta h = (2.0)(10)(3.0) = 60 \text{ J}$. 6.0: forgetting g . 5.0: adding instead of multiplying.

Exercise 3 — Energy stores (3 marks, 3 🍷)

Solution — A ball is thrown upwards, while it rises: kinetic $\rightarrow$ gravitational potential; A compressed spring in a toy dart gun is released: elastic (strain) $\rightarrow$ kinetic; A diver climbs the ladder to the top board: chemical $\rightarrow$ gravitational potential. Award the mark only if both stores are right.

Exercise 4 — Two travellers (8 marks, 2 🍷)

Solution — (a) 🍷 [2] The cyclist is at rest for 1 h (horizontal line from 0 to 1 h). The graphs cross at $t = 1.5 \text{ h}$, 9 km from P. 1 mark each.

(b) [2] While moving, from 1 h to 2 h, the distance from P changes by 18 km in 1 h. The line goes up because the cyclist is moving away from P.

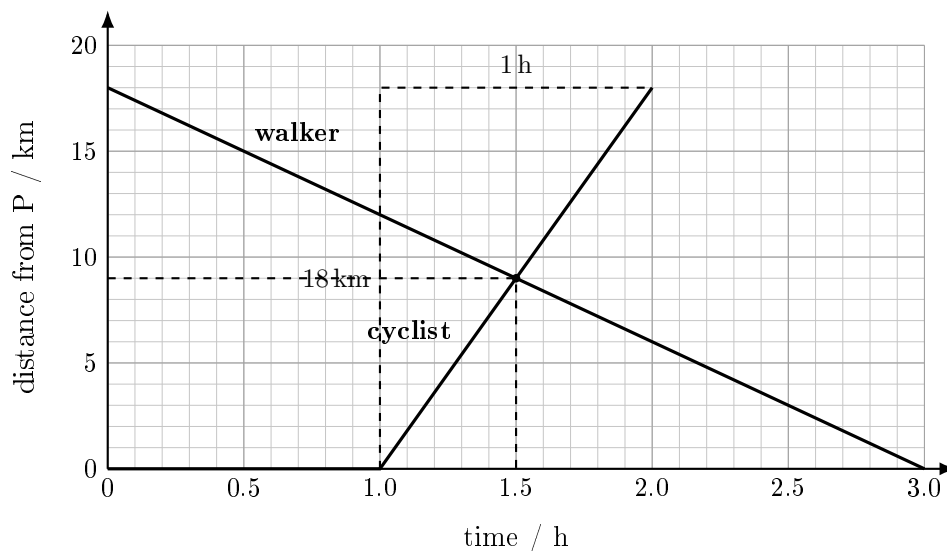
$$v = \frac{s}{t} = \frac{18 \text{ km}}{1 \text{ h}} = 18 \text{ km/h}$$

(c) [2] The cyclist covers 18 km in the whole time 2 h, including the rest.

$$v_{\text{av}} = \frac{18 \text{ km}}{2 \text{ h}} = 9 \text{ km/h}$$

(d) [2] The walker covers 18 km in 3 h, so $v = 6 \text{ km/h}$.

$$v = \frac{6000 \text{ m}}{3600 \text{ s}} \approx 1.67 \text{ m/s}$$



Check and common mistakes — (a) Writing 1.5 h for the rest (that is the time of the meeting) or giving the meeting point in hours instead of kilometres. (b) Using the whole time 2 h gives 9 km/h: the rest is not moving time. Dividing time by distance gives 0.056: the unit would be h/km. (c) Using 1 h gives 18 km/h: the rest counts in the average. Averaging 0 and 18 gives 9 km/h: wrong. (d) Multiplying by 3.6 instead of dividing gives 21.6 m/s: not a possible speed for a walker. Check: $1.67 \text{ m/s} \times 3.6 \approx 6 \text{ km/h}$.

Exercise 5 — A motorbike brakes (6 marks, 2 🎯)

Solution — (a) 🎯 [1] Thermal (internal) energy of the brakes, the tyres and the road.

(b) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(250)(20)^2 = 50\,000 \text{ J}$$

(c) [2] The work done by the brakes equals the kinetic energy lost: $Fd = E_k$ (1 mark).

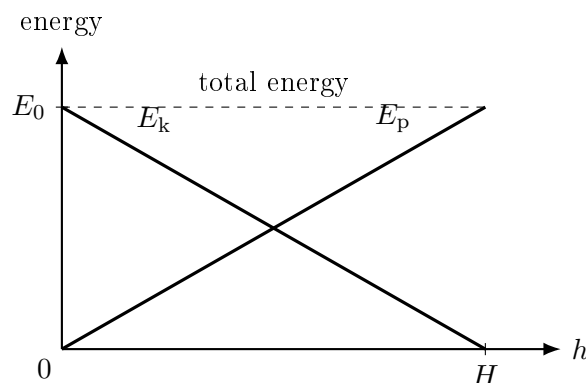
$$F = \frac{E_k}{d} = \frac{50\,000}{40} = 1250 \text{ N}$$

(d) 🎯 [1] Half as long: the kinetic energy to remove is the same and, with a force twice as big, $d = E_k/F$ is halved. Check: $d = 20 \text{ m}$.

Check and common mistakes — (b) Forgetting the square gives 2500 J. Forgetting the $\frac{1}{2}$ gives 100 000 J. (c) With 2500 J from (b) the force would be 62.5 N. Dividing by the mass gives 200: that is not a force (the unit would be J/kg). (d) “One quarter as long” is the answer for a halved speed, not for a doubled force.

Exercise 6 — A tennis ball thrown upwards (10 marks, 6 🎯)

Solution — (a) 🎯 [2]



The kinetic energy decreases linearly from E_0 to 0 (1 mark). The potential energy increases linearly from 0 to E_0 (1 mark). The two lines cross at $H/2$ and the total stays constant.

(b) [2]

$$\frac{1}{2}mv^2 = mgH \Rightarrow H = \frac{v^2}{2g}$$

1 mark for the equality, 1 for the final expression. The mass cancels.

(c) [2]

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(0.06)(20)^2 = 12 \text{ J}$$

(d) [2]

$$H = \frac{E_k}{mg} = \frac{12}{(0.06)(10)} = 20 \text{ m}$$

(e) [2] Four times as big (1 mark): $H \propto v^2$, so doubling v multiplies H by 4 (1 mark). Check: $H = 80 \text{ m}$.

Check and common mistakes — (c) Forgetting the square gives 0.6 J. (d) Forgetting g gives 200 m. Check with (b): $H = v^2/2g = 20^2/20 = 20 \text{ m}$.

Exercise 7 — The half-pipe (8 marks, 4 [2])

Solution — (a) [2] She rises to 4.0 m above C, the height of A (1 mark). At A she has only gravitational potential energy; no energy is lost, so at the highest point, where she stops for an instant, she again has only gravitational potential energy and the total is the same (1 mark). Accept 3.2 m above E if it is clearly stated as the height above E.

(b) [2]

$$mgh_A = mgh_E + \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2g(h_A - h_E)}$$

1 mark for the energy balance, 1 for the final expression. The mass cancels.

(c) [2]

$$\Delta E_p = mg\Delta h = (50)(10)(4.0 - 0.8) = (50)(10)(3.2) = 1600 \text{ J}$$

(d) [2] The potential energy lost becomes kinetic energy:

$$\begin{aligned} \frac{1}{2}mv^2 &= 1600 \Rightarrow v^2 = \frac{2 \times 1600}{50} = 64 \\ v &= \sqrt{64} = 8.0 \text{ m/s} \end{aligned}$$

Check and common mistakes — (a) 3.2 m is the height above the lip, not above C. (c) Using h_A alone gives 2000 J; using h_E alone gives 400 J: the energy lost depends on the difference. Forgetting g gives 160 J. (d) Forgetting the $\frac{1}{2}$ gives $v^2 = 32$, $v \approx 5.7 \text{ m/s}$. Forgetting the square root gives 64 m/s. Check with (b): $\sqrt{2 \times 10 \times 3.2} = 8.0 \text{ m/s}$.

Exercise 8 — Lifting and carrying a box (8 marks, 4 [2])

Solution — (a) [2] $W = Fd$ (work done by the hands, with d the distance moved in the direction of the force) and $\Delta E_p = mg\Delta h$. 1 mark each.

(b) [2]

$$\begin{aligned} W &= Fd = (100)(0.90) = 90 \text{ J} \\ \Delta E_p &= mg\Delta h = (8.0)(10)(0.90) = 72 \text{ J} \end{aligned}$$

(c)  [1] The difference is $90 - 72 = 18 \text{ J}$. It is the kinetic energy of the box, which is still moving at the height of a table.

(d)  [1] 0 J.

(e) [2] The force of the hands is vertical and the box moves horizontally: the box does not move in the direction of the force, so $d = 0$ in $W = Fd$ and $W = 0$ (1 mark). The box gains no energy: its height and speed do not change (1 mark). She still uses chemical energy in the muscles, but that is not work done on the box.

Check and common mistakes — (b) Using the weight $mg = 80 \text{ N}$ instead of the force of 100 N gives $80 \times 0.90 = 72 \text{ J}$, equal to the gain in E_p : the difference disappears. Forgetting g in ΔE_p gives $8.0 \times 0.90 = 7.2 \text{ J}$. (d) Using 15.0 m gives $80 \times 15.0 = 1200 \text{ J}$: that is the distance moved, but not in the direction of the force. (e) Answers like “carrying costs no energy” are wrong: the muscles transfer chemical energy to thermal energy even when no work is done on the box.